



Society of Petroleum Engineers

SPE-227612-MS

Data-Centric Approach: Digital Innovation for Elevating Casing Exit Planning and Execution

Hany Gamal, Weatherford Saudi Arabia

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227612-MS](https://doi.org/10.2118/227612-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Casing exit operations, essential for accessing additional reservoirs or bypassing wellbore obstructions, demand meticulous planning, accurate whipstock placement, and efficient execution. This process often involves challenges related to wellbore complexity and variable downhole conditions. A data-centric approach, leveraging digital innovation, introduces transformative solutions for optimizing planning, execution, and overall performance. This study investigates how advanced data management, real-time analytics, and digital decision-support systems elevate the efficiency, reliability, and performance of casing exit operations, driving operational success.

The study proposes a data-centric methodology workflow that integrates advanced data management with a digital platform to enhance decision-making and operational execution in casing exit operations. The workflow begins with the analysis of historical job data to identify patterns, benchmarks, and best practices for whipstock placement and milling efficiency. It focuses on deep sensitivity analytics, considering surface rig sensor data (e.g., weight on mill, hook load, drilling torque, drillstring rotation speed, pumping rate, and pressure) in conjunction with window milling and rathole drilling progress, particularly rate of penetration (Gamal et al., 2023). Additionally, it takes into account the downhole environment, including formation type and strength, as well as casing and cement integrity.

The data-centric methodology implemented in casing exit operations has significantly improved key performance indicators (KPIs), enhancing both operational efficiency and risk mitigation. The deployment of interactive data analytics and visualization dashboards, engineers gained real-time, actionable insights, allowing dynamic adjustments to optimize both the planning and execution phases. The integrated digital platform facilitated seamless communication and alignment among all teams, enabling faster decision-making and improving collaboration throughout the operation. Key improvements include 10 to 15% reduction in operational time with faster execution and improved efficiency. Whipstock placement accuracy was also enhanced, as data-driven insights ensured better precision, reducing rework and improving access to targeted reservoir zones. Furthermore, there was a noticeable reduction in non-productive time (NPT), as dynamic data adjustments minimized equipment failures and misalignments, ensuring smoother operations. The engineering workflow also resulted in a reduction in risk exposure, with predictive analytics identifying potential failures early, enabling proactive mitigation, and avoiding costly delays. After all, the

approach increased operational predictability, with advanced data analytics ensuring that drilling operations stayed within optimal limits, reducing unanticipated costs. The field implementation of this data-driven approach demonstrated its value, boosting efficiency, mitigating risks, and lowering operational costs. By comparing real-time performance data with historical benchmarks, operators were able to fine-tune operations dynamically, achieving superior results with fewer resources.

The proposed engineering approach represents a paradigm shift in casing exit operations by seamlessly integrating digital innovation into every phase of the process, from planning to execution. Key novel contributions include proactive risk mitigation through advanced data analytics, predictive performance insights driven by real-time data, and enhanced collaboration across teams facilitated by digital platforms. These innovations empower the engineering team to proactively address challenges, optimize operational efficiency, and consistently achieve superior results. By adopting a data-centric methodology, this approach significantly elevates the intelligence, efficiency, and reliability of casing exit operations, setting a new industry standard and redefining operational excellence.

Key Words: Data management, digital innovation, real-time analytics, casing exit, operational efficiency, planning optimization, data-centric workflow, operation dashboards

Introduction

Casing exit operations, critical interventions for accessing bypassed reserves, or sidetracking past wellbore impediments, represent a fundamental yet technically demanding process in wellbore construction and intervention. Successfully creating a precise window through existing casing and cement, followed by accurate rathole drilling into a new formation, requires meticulous pre-job planning, accurate tool deployment, and highly controlled real-time execution (Vuyk, 2016; Gamal et al., 2024b). These operations are inherently susceptible to challenges arising from the variable and often complex condition of the existing wellbore, the properties of the cement sheath, and the unpredictable nature of downhole conditions encountered during milling and drilling (Figure 1). Suboptimal execution or unforeseen issues during a casing exit can lead to significant operational inefficiencies, including excessive operational time, increased non-productive time (NPT), potential damage to the existing casing string, and even sidetrack failure, resulting in costly delays and remedial operations.

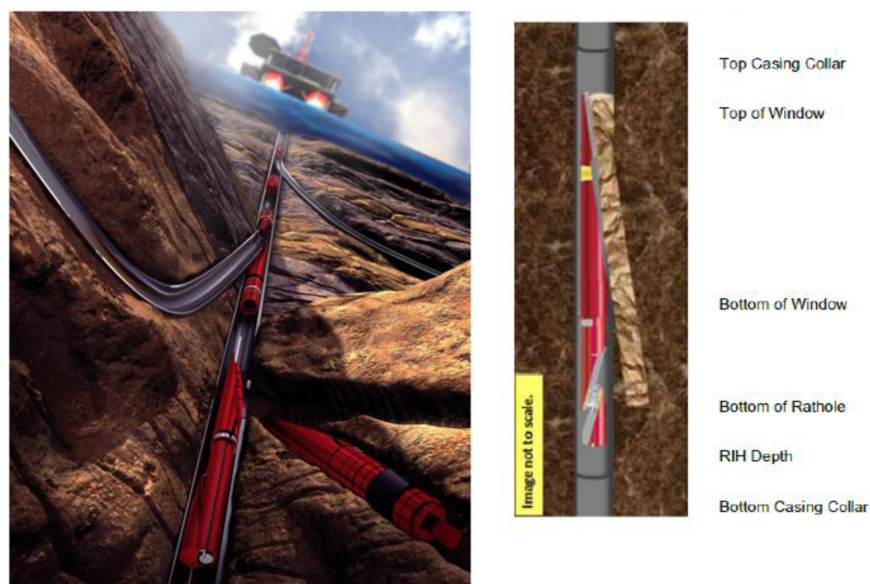


Figure 1—Casing Exits for Multilaterals

Historically, planning and executing the casing exit operations have relied heavily on accumulated experience, offset well data review (often limited in real-time detail), and manual adjustments based on surface parameter monitoring. While effective to a degree, this traditional approach often lacks the precision and adaptability required to consistently achieve optimal results in challenging environments. The limitations become particularly apparent when dealing with complex wellbore geometries, uncertain cement quality, or formations with variable drillability behind the casing.

The rapid advancements in digital technologies, coupled with the increasing availability of high-frequency operational data from modern drilling and intervention rigs, present a transformative opportunity to revolutionize casing exit operations (Trindade et al., 2023). A data-centric approach, one that places the collection, integration, analysis, and utilization of data at the core of every phase of the casing exit process, unlocks unprecedented potential for enhancing efficiency, reliability, and overall performance. By leveraging the power of digital innovation through encompassing advanced data management, real-time operational analytics, sophisticated modeling, and integrated digital decision-support systems, engineering teams can gain profound insights into downhole conditions, predict performance, proactively identify potential issues, and make dynamic adjustments that were previously not possible (Febowitz, 2013; Rossi et al., 2018; Gamal et al., 2024a).

Technical Challenges for Casing Exit Operations

Casing exit operations, as a specialized well intervention, face a unique set of technical challenges that differentiate them from conventional drilling. These challenges are primarily related to working within an existing wellbore and interacting with the casing and cement sheath. Figure 2 presents a technical breakdown of the key factors that collectively contribute to a successful casing exit window, organized into three primary categories: pre-job planning and design, wellbore and geological conditions, and execution and operational control.

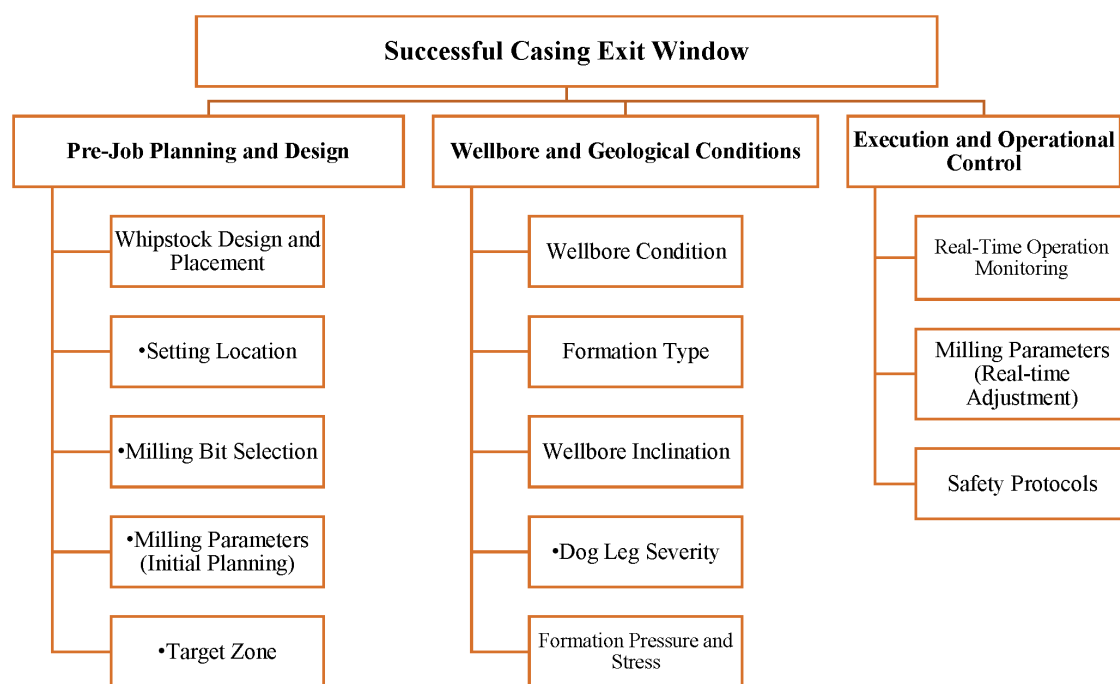


Figure 2—Technical Factors Governing Casing Exit Window Performance

Pre-job planning and design phase considers the technical decisions made during this phase directly impact the feasibility and efficiency of the operation. Critical challenges include precise whipstock design

and placement. This involves selecting a whipstock system capable of withstanding anticipated loads and designing its deployment and setting procedure to achieve the intended sidetrack trajectory. Casing exit operations utilize distinct system designs, including 2-degree and 3-degree configurations as shown in Figure 3. The 2-degree system offers high-performance, single-trip milling, gauge retention, and formation drilling, often with a retrievable premium anchor for zonal isolation or abandonment. It's suited for full-bore laterals, extended reach, multilateral completions, and multi-string exits. The 3-degree system provides a single-trip, full-gauge milling capability with a single-angle concave whipstock and versatile anchor options (mechanical/hydraulic), designed for a smooth transition into the lateral, adaptable for short-radius or extended reach wells.



Figure 3—Whipstock Casing Exit Systems

Determining the optimal setting location requires considering factors like casing condition, cement quality, and the desired sidetrack starting point, often complicated by uncertainties in wellbore surveys and friction effects on depth correlation. Milling bit selection is a technical challenge requiring careful consideration of the casing material properties (grade, thickness), the expected hardness of the cement and formation behind casing, and the desired window geometry to select a mill type and size that can efficiently cut the window without premature wear or damage. Milling parameters (initial planning) involve predicting the appropriate ranges for weight on mill (WOM), rotary speed (RPM), and flow rate based on available data and models, recognizing that these initial parameters may require dynamic adjustment during execution due to downhole variability. Finally, accurately defining the target zone translates into technical requirements for whipstock orientation and trajectory control during rathole drilling to ensure the new wellbore intersects the intended geological objective.

Wellbore and geological conditions consider these inherent subsurface factors present technical constraints and uncertainties that must be understood and managed. The wellbore condition, particularly the state of the existing casing and cement at the planned exit point, is a fundamental challenge. Pre-existing casing wear, deformation, or varying cement bond quality introduce technical complexities that impact milling difficulty, the stability of the casing during milling, and the support provided by the formation/cement behind the casing. The formation type and its mechanical properties (strength, abrasivity)

immediately behind the casing and in the rathole section present technical uncertainties that influence milling and drilling parameters, rate of penetration (ROP), and bit/mill wear. Wellbore inclination and dog leg severity define the directional geometry of the existing wellbore. Technically, these parameters influence the forces (torque and drag) experienced during tool run-in and retrieval, the ability to maintain tool face orientation during whipstock setting and rathole drilling, and the potential for differential sticking or hang-up. Formation pressure and stress represent the in-situ geomechanical environment. Technically, these factors influence the stability of the open-hole rathole section and the risk of fluid loss or influx during drilling, requiring careful management of drilling fluid properties.

The execution and operational control category focus on the technical challenges of managing the operation in real-time. Real-time operation monitoring is technically challenging due to the need to acquire, transmit, and process high-frequency data streams from numerous surface and potentially downhole sensors to accurately assess operational performance and detect subtle anomalies indicative of potential issues. Milling parameters (real-time adjustment) present the technical challenge of dynamically optimizing parameters such as WOM, RPM, and flow rate during window milling and rathole drilling based on real-time data feedback (e.g., ROP, Torque, Vibration, MSE). This requires the ability to interpret real-time data patterns and make informed adjustments to maximize efficiency and avoid detrimental operating conditions like severe vibration or stick-slip (Saadeldin et al., 2023). Maintaining strict adherence to safety protocols is a fundamental technical and procedural requirement throughout the operation to mitigate the inherent risks associated with working with high pressures, rotating machinery, and downhole tools in potentially unstable wellbore environments.

Addressing these interwoven technical challenges through robust planning, a thorough understanding of wellbore and geological conditions, and vigilant real-time operational control is essential for mitigating the cumulative risk of NPT and sidetrack failure, ultimately leading to the successful creation of the casing exit window.

Data-Centric Methodology Workflow for Casing Exit

The proposed data-centric methodology workflow integrates advanced data management, real-time analytics, and digital decision support systems throughout the casing exit operation lifecycle to proactively address the specific technical challenges. This approach shifts from a reactive model to one that leverages data for informed planning, precise execution, and continuous optimization (Figure 4).



Figure 4—Technical Workflow and Engineering Methodology

Phase 1: Comprehensive Pre-Job Data Integration

The initial and foundational phase of the data-centric methodology for optimizing casing exit operations is comprehensive pre-job data integration and planning. This phase is critical for establishing the necessary data backbone and leveraging a wide array of historical and well-specific information to develop a robust and optimized casing exit plan. It begins with the meticulous integration of disparate data sources, where all relevant historical and current well data are aggregated into a unified digital platform. This includes detailed historical casing exit job data from previous operations in the field. Analysis of this historical data goes beyond simple summary statistics; it focuses on identifying statistical distributions of operational times for specific activities like whipstock setting, window milling, and rathole drilling, uncovering common failure modes and their technical triggers (e.g., hook load analysis, correlation between torque/drag profiles and running issues, identification of specific vibration patterns associated with milling problems), and analyzing

the range of operational parameters (WOM, RPM, flow rate) used in successful past operations to define effective operating envelopes.

Crucially, well-specific construction data is integrated, encompassing original drilling reports, detailed casing manifests providing technical specifications (size, weight, grade, connection type), comprehensive cementing reports (slurry design properties, volumes pumped, pressures), and any available diagnostic logs such as cement bond logs (CBL/VDL) or ultrasonic imaging tools (USIT). This data provides essential technical information about the physical properties and condition of the casing and cement sheath at the planned exit point. Wellbore survey data, including comprehensive directional surveys (MWD, Gyro), is integrated to accurately define the existing wellbore trajectory at and above the planned exit depth; pre-job high-accuracy gyro surveys are particularly essential for providing a reliable technical reference for precise whipstock orientation. Finally, detailed equipment specifications for the chosen whipstock system, milling assembly, and drill bit, including dimensions, material properties, and operating limits, are integrated into the planning platform. Building upon this integrated data, performance benchmarking from historical data is conducted. This analysis moves beyond simple average performance to identify top-quartile performance metrics for key casing exit activities (e.g., the best time achieved to mill a foot of casing of a specific specification, the most efficient rathole ROP recorded in a particular formation). These top-quartile metrics technically represent achievable performance targets and serve as realistic benchmarks for the optimized execution phase.

Phase 2: Simulation and Sensitivity Analytics

Phase 2 focuses on pre-job simulation and sensitivity analytics to optimize the casing exit plan and anticipate issues. Hydraulic modeling is critical for efficient hole cleaning and pressure management (Alharbi et al., 2025). Figure 5 illustrates hookload simulations and actual measurements for analyzing drag and friction. Multiple lines represent simulated hookloads under various friction factors (FF), as the solid black line represents the hookload as a reference, while scattered points show the actual measured hookload. Deviations could indicate changes in wellbore conditions, fluid properties, or sticking issues.

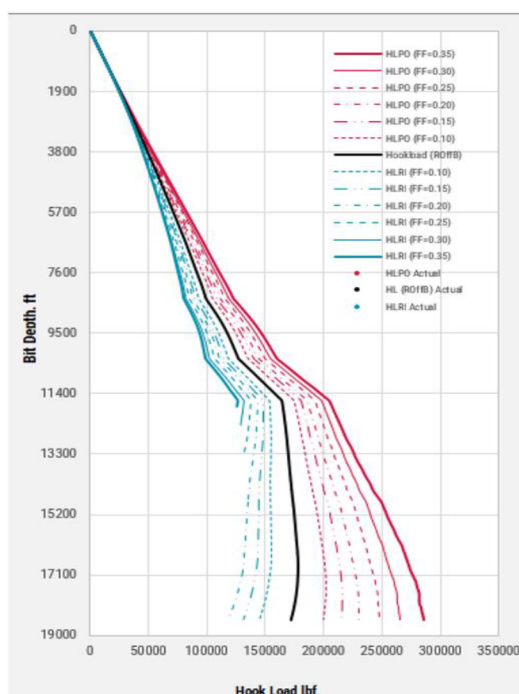


Figure 5—Simulated Hookloads Under Various Friction Factors

Detailed sensitivity analytics vary input parameters within the platform engineering models to understand their impact on ROP, vibration, torque, and tool performance, and evaluate sensitivity of whipstock orientation and BHA influence on trajectory. Based on these results, optimized operational parameters and procedures are defined, including recommended ranges for setting force, WOM, RPM, and flow rate for milling and rathole drilling. refined performance benchmarks (e.g., target milling time per foot, rathole ROP) and clear operating limits and technical thresholds for key parameters (torque, vibration, hookload) are established for real-time monitoring during execution.

Phase 3: Real-Time Data Acquisition

During the actual casing exit operation, the digital platform continuously acquires and integrates high-frequency real-time data from multiple technical sources. A digital platform serves data management and real-time monitoring throughout the workflow. Data acquisition and aggregation system to capture high-frequency real-time data from all relevant surface sensors (hookload, torque, RPM, pump rate, standpipe pressure, flow in/out) and downhole sensors if available, and streaming the data into a unified database within the digital platform with advanced technological capabilities for optimizing the operation performance (AlBar et al., 2018; Gamal, 2025). Creating intuitive dashboards within the digital platform to provide real-time visibility of key performance indicators (KPIs), fluid properties, and operational parameters for the engineering and operations teams (Ferrari et al., 2021). Figure 6 represents a synthetic sample from the real-time drilling digital platform, providing real-time operational data and performance analysis.

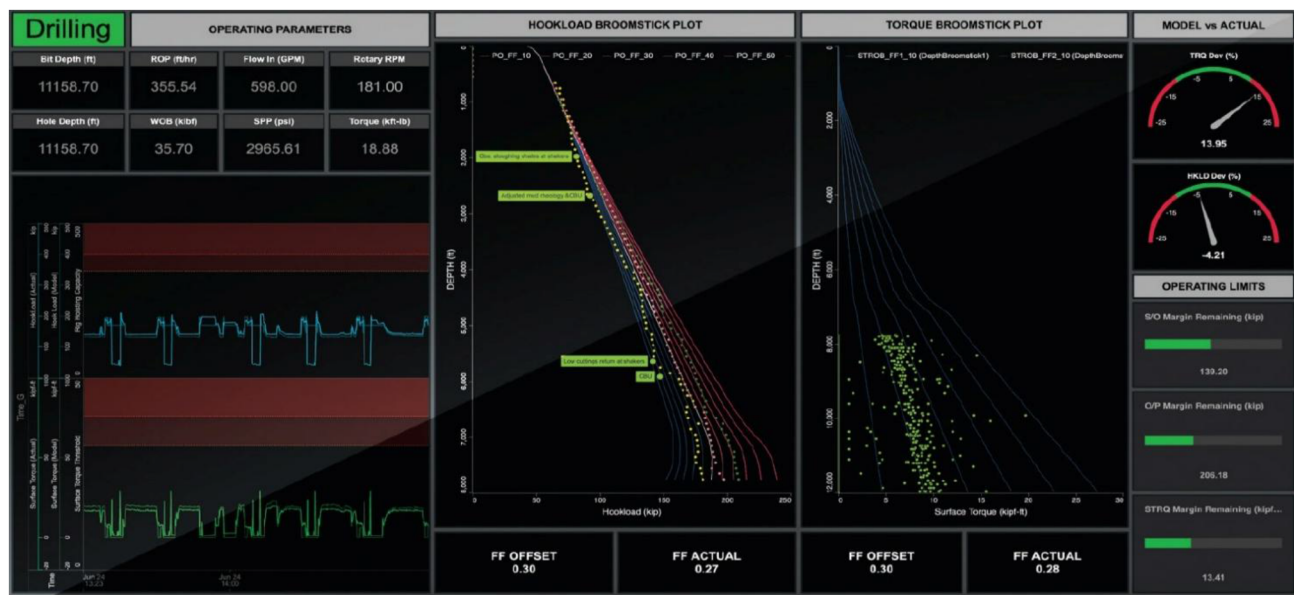


Figure 6—Drilling Digital Platform (Whitfield, 2022)

The data analytics visually analyze the rate of penetration ROP relative to key milling/drilling parameters, categorized by geological formation, enabling direct comparative technical assessment across different operational ranges as presented in Figure 7. The initial panel displays kernel density estimates for ROP, revealing distinct performance distributions for each 'Formation'. For instance, 'Formation E' shows a higher ROP peak, indicating greater drillability compared to 'Formation D' or 'Formation C', which exhibit lower and broader ROP distributions. Subsequent scatter plots detail bivariate relationships, with ROP against normalized WOB, Torque, RPM, SPP, and Flow, color-coded by 'Formation'. This stratification is crucial for engineering analysis as the ROP vs. WOB plot highlights how varying formations respond to applied weight, potentially guiding optimal WOB selection. Similarly, ROP relationships with Torque,

RPM, SPP, and Flow pumps offer insights into mechanical efficiency, hydraulic optimization, and the dynamic interaction between the bit and formation. This comprehensive visual provides a rapid, data-driven understanding of performance variations influenced by geological heterogeneity, supporting informed parameter adjustments and operational strategy optimization for enhanced drilling efficiency.

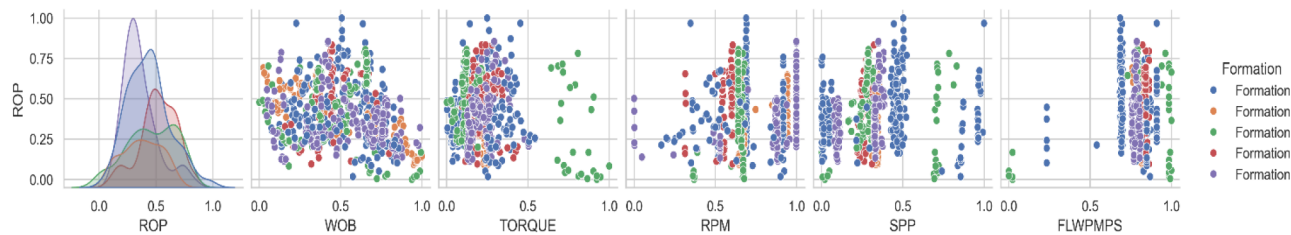


Figure 7—Data Analytics

Downhole directional measurements (tool face, inclination, azimuth) are essential for real-time monitoring of whipstock orientation during deployment and trajectory control during rathole drilling. Reliable and low-latency transmission of all this real-time data from the rig to the central digital platform is ensured. Robust data processing algorithms are implemented for real-time cleaning, filtering, aligning, and synchronizing these high-frequency data streams to maintain accuracy for subsequent analysis and decision-making.

Phase 4: Dynamic Optimization

This phase represents the technical apex of the data-centric approach during casing exit execution. This phase is fundamentally driven by continuous real-time data analysis, which yields actionable insights directly supporting dynamic decision-making on the rig floor. A core function within this phase is the real-time key performance indicators generation and visualization, specifically tailored for the casing exit operations. The system continuously calculates and displays these KPIs on interactive dashboards. These KPIs are technically derived by processing the integrated high-frequency surface and downhole data streams, providing immediate, quantitative feedback on the operational performance of the casing exit process. Empowered by these real-time analytics and the integrated decision support, operational personnel can make dynamic adjustments to parameters and procedures. This includes real-time parameter optimization for both milling and drilling, where parameters such as WOM, RPM, and flow rate are adjusted in real-time. The technical goal of these adjustments is to maintain optimal performance, minimize tool wear, and ensure a stable and efficient operation.

Figure 8 illustrates a systematic approach to adjusting drilling parameters over time based on the real time data analytics, aimed at optimizing operational performance. It depicts discrete time intervals where "crew action" consistently involves "Optimize Parameters," signifying continuous operational refinement. The associated parameter adjustments for "Optimized RPM", "Optimized WOM," and "Optimized Flow" reveal a strategic adaptation. Specifically, RPM and Flow are generally increased, indicating a drive for enhanced energy delivery and hydraulics. WOM, however, undergoes intermittent adjustments, including strategic decreases or periods of "No Change," often reflecting efforts to manage downhole dynamics, control vibrations, or optimize ROP within specific drilling conditions.

The dynamic optimization demonstrates a real-time, data-driven optimization loop where milling/drilling parameters are refined based on ongoing feedback, aiming to achieve and sustain peak operational efficiency, as opposed to static parameter application.

Dynamic Optimization				
TIME	Crew Action	Optimized RPM	Optimized WOM	Optimized Flow
1.0	Optimize Parameters		No Change	
2.0	Optimize Parameters			
3.0	Optimize Parameters		No Change	
4.0	Optimize Parameters			
5.0	Optimize Parameters			
6.0	No Action	No Change	No Change	No Change

Figure 8—Dynamic Optimization

Phase 5: Post-Job Analysis

This phase involves a comprehensive technical analysis utilizing the archived operational data to meticulously evaluate performance, systematically capture lessons learned, and enrich the organizational knowledge base for the planning and execution of future operations. A key activity is detailed performance evaluation, which involves analyzing the high-frequency real-time data recorded throughout the operation to precisely quantify actual performance metrics for each distinct stage of the casing exit process. This includes metrics such as whipstock run-in and setting time, total time spent on window milling, the time taken to mill each foot of casing, the total rathole drilling time, and the average and instantaneous rathole ROP. These quantified metrics are then technically compared against the planned benchmarks established during the pre-job phase and against historical performance data from similar offset operations. Furthermore, root cause analysis of events is conducted, utilizing the detailed real-time data streams and synchronized daily drilling reports to perform in-depth technical investigations into any operational issues, deviations from the plan, or NPT events that occurred, regardless of their reported severity. Documentation of lessons learned and best practices is formally undertaken, capturing key technical observations, successful strategies employed, optimized operational parameters identified through real-time analysis, and any valuable insights gained during both the planning and execution of the casing exit operation. This knowledge is technically captured and stored within the digital platform's knowledge management system, making it accessible for future reference. This comprehensive dataset technically enriches the historical dataset available for future pre-job planning, performance benchmarking, and the training and validation of predictive models, thereby creating a continuous technical feedback loop essential for the iterative improvement of the data-centric casing exit workflow.

Operational Outcomes

The platform's primary benefit was fostering seamless communication and alignment among all teams, which significantly streamlined workflows, enabled faster decision-making, and enhanced overall collaboration throughout the operation. Quantifiable improvements include a 10 to 15% reduction in total operational time, attributed to faster execution and improved efficiency across various phases of the casing exit process. A key technical outcome was the enhanced whipstock placement accuracy; data-driven insights provided by the platform ensured greater precision in positioning and orienting the whipstock, directly

reducing the need for costly rework and improving access to the intended reservoir zones. Furthermore, the platform contributed to a noticeable reduction in non-productive time (NPT). This was achieved through dynamic data adjustments and real-time monitoring capabilities, which helped minimize equipment failures and misalignments, thereby ensuring smoother and more reliable casing exit operations.

Conclusions

- This study investigates the profound impact of implementing a data-centric approach, facilitated by a digital platform, specifically on elevating the planning and execution of casing exit operations.
- The data-centric approach presented in this study offers a transformative solution through leveraging advanced data management, real-time analytics, and digital decision-support systems. Engineering teams can significantly elevate the efficiency, reliability, and performance of these critical well interventions.
- The proposed methodology workflow, which begins with the comprehensive analysis of historical casing exit job data and well-specific construction details, integrates real-time data acquisition from rig sensors and downhole tools, employs advanced analytics for generating actionable insights, and facilitates dynamic adjustments during execution via a central digital platform, provides a systematic and robust framework specifically tailored for optimizing casing exit operations.
- Quantifiable improvements were demonstrated across key performance indicators directly relevant to casing exit success, including a significant 10 to 15% reduction in overall operational time, enhanced whipstock placement accuracy, and a noticeable reduction in non-productive time.
- The approach also contributed to a reduced risk exposure through proactive identification of potential issues and increased operational predictability by comparing real-time performance to benchmarks and predictive models.
- Crucially, the establishment of this robust, data-centric framework and the consistent generation of high-quality, integrated operational data lay a vital foundation and implementation of future machine learning models. This structured dataset provides the necessary training data for predictive analytics, automated parameter optimization, and advanced anomaly detection, further enhancing the intelligence and automation of casing exit operations.

Acknowledgment

The author expresses sincere appreciation to Weatherford Company for its invaluable support in conducting this study and for granting publication permission.

References

1. AlBar, A.H., Alotaibi, B.M., Asfoor, H.M., Nefai, M.S., 2018. A Journey Towards Building Real-Time Big Data Analytics Environment for Drilling Operations: Challenges and Lessons Learned. Presented at the SPE Kingdom of Saudi Arabia Annual Technical Symposium and Exhibition, OnePetro. <https://doi.org/10.2118/192285-MS>
2. Alharbi, A., Petre, R., Gamal, H., 2025. Casing Exit Excellence: a Holistic Strategy of Integrated Engineering Models and Well Integrity Analytics for Extended Reach Wells.
3. Feblowitz, J., 2013. Analytics in Oil and Gas: The Big Deal About Big Data. Presented at the SPE Digital Energy Conference, OnePetro. <https://doi.org/10.2118/163717-MS>
4. Ferrari, F., Naselli, R., Brunetti, P., Michelez, J., Zini, E., 2021. Digitalization for Reducing Carbon Footprint in Drilling Operations. Presented at the Abu Dhabi International Petroleum Exhibition & Conference, OnePetro. <https://doi.org/10.2118/207407-MS>

5. Gamal, H., 2025. Technological Integration and Innovative Strategies Harnessing Artificial Intelligence for Operational Excellence [WWW Document]. <https://services.igi-global.com/resolvedoi/resolve.aspx?doi=10.4018/979-8-3693-6447-5.ch009>. URL <https://www.igi-global.com/gateway/chapter/www.igi-global.com/gateway/chapter/359302> (accessed 5.29.25).
6. Gamal, H., Albahri, S., Essa, H., Srivastava, M., Alghamdi, N., Shaikh, W., 2024a. Driving Operational Excellence: Cost Savings and Performance Optimization Through Tripping and Connection Time Optimization.
7. Gamal, H., Elkatatny, S., Al Gharbi, S., 2023. Rig Sensor Data for AI-ML Technology-Based Solutions: Research, Development, and Innovations. Presented at the ADIPEC, OnePetro. <https://doi.org/10.2118/216429-MS>
8. Gamal, H., Gomez, H., Alshalan, M., Alshaarawi, A., 2024b. Successful Field Trials for Tracking Casing Exit: Innovative Sensing Technology Application, in: *Day 2 Tue, February 13, 2024*. Presented at the International Petroleum Technology Conference, IPTC, Dhahran, Saudi Arabia, p. D021S061R002. <https://doi.org/10.2523/IPTC-23750-MS>
9. Rossi, N., Michelez, J., Concina, F., 2018. Technology Update: Big Data for Advanced Well Engineering Holds Strong Potential To Optimize Drilling Costs. *Journal of Petroleum Technology* **70**, 18–21. <https://doi.org/10.2118/0218-0018-JPT>
10. Saadeldin, R., Alsabaa, A., Gowida, A., Gamal, H., Elkatatny, S., 2023. Machine Learning Advisory System for Mitigating Downhole Vibrations for Horizontal Sections. Presented at the ADIPEC, OnePetro. <https://doi.org/10.2118/216123-MS>
11. Trindade, D.N., Duarte, L.G., Perico, I., Bandeira, G.L., 2023. Driving Change in the Oil and Gas Industry: A. Digital Transformation Framework. Presented at the Offshore Technology Conference Brasil, OnePetro. <https://doi.org/10.4043/32860-MS>
12. Vuyk, A., 2016. Remote Collaboration Reduces Risk in Casing-Exit Jobs Around the Globe. Presented at the SPE Annual Technical Conference and Exhibition, OnePetro. <https://doi.org/10.2118/181458-MS>
13. Whitfield, S., 2022. The art of data science: Industry hones in on reliability, ease of use. *Drilling Contractor*. URL <https://drillingcontractor.org/the-art-of-data-science-industry-hones-in-on-reliability-ease-of-use-63276> (accessed 6.11.25).